

Mohan Sai Krishna Guntupalli

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Professional Summary:

- I am a Data Engineer with around 9+years of experience in Cloud (AWS, Azure), Data warehousing, Data engineering, Feature engineering, Hadoop big data, **ETL/ELT**, and Business Intelligence. As a Cloud data architect and engineer, specializing in **AWS** and **Azure frameworks, Cloudera, Hadoop Ecosystem, Spark/Py Spark/Scala, Data bricks, Hive, Redshift, Snowflake**, relational databases, tools like **Tableau, Airflow, DBT, Presto/Athena, Data DevOps Frameworks/Pipelines Programming skills in Python**.
- Experience in building data solutions using SQL Server, MSBI and AWS, Azure Cloud
- Comprehensive experience in BI & OLAP tools like IBM Cognos Analytics (C11)/Cognos BI 10.2.2 (Report Studio, Framework Manager, Report Studio, Workspace advanced, Analysis Studio, Query Studio, Event Studio, Cognos Connection, Power Play Transformer), Cognos Planning 8.3, and ETL tools Informatica, and IBM Data Manager/Decision Stream.
- Worked on Azure Cloud, Azure Data Factory, Azure Data Lake Storage, Azure Synapse Analytics, Azure Analytical services, Azure Cosmos NO SQL DB, and Data bricks.
- Experience with an in - depth level of understanding in the strategy and practical implementation of AWS Cloud Specific technologies including **EC2, EBS, S3, VPC, RDS, SES, ELB, EMR, ECS, Cloud Front, Cloud Formation, Elastic Cache, Cloud Watch, Red Shift, Lambda, SNS, Dynamo DB, Kinesis**.
- Design and implement Infrastructure as Code (IaC) using Terraform and other automation tools.
- Hands-on experience on AWS cloud services (**VPC, EC2, S3, RDS, Redshift, Data Pipeline, EMR, DynamoDB, Workspaces, Lambda, Kinesis, RDS, SNS, SQS**)
- Experience in Hadoop Ecosystem components like Hive, HDFS, Sqoop, Spark, Kafka, Pig.
- Experience in architecting, designing, installation, configuration, and management of Apach e Hadoop Clusters, MapR, Horton works & Cloudera Hadoop Distribution.
- Good understanding of Hadoop architecture and Hands on experience with Hadoop components such as Resource Manager, Node Manager, Name Node, Data Node and Map Reduce concepts and HDFS.
- Expertise in Data Migration, Data Profiling, Data Ingestion, Data Cleansing, Transformation, Data Import, and Data Export using multiple ETL tools such as Informatica Power Centre.
- Effectively implemented performance tuning techniques in **Redshift**, including optimization through Distribution Keys, Sort Keys, and Partitioning. Additionally, proficiently managed user access and permissions to AWS resources utilizing **IAM** and transition existing AWS infrastructure into Infrastructure as Code (**IaC**) using **Terraform**.
- Hands on Spark RDD, Data frame API, Data set API, Data Source API, Spark SQL, and Spark Streaming.
- Adept at utilizing BI tools such as Power BI and QlikView for enhancing reporting capabilities and developing BI applications in accordance with client requirements. knowledge in Data Modelling and Data analyzing in MS SQL Server and in query optimization.
- Extensive experience in developing complex Stored Procedures, Functions, Triggers, Views, Cursors, Indexes, CTE's, Joins and Sub queries with T-SQL.
- Experienced in managing Azure Data Lakes (ADLS) and Data Lake Analytics and an understanding of how to integrate with other Azure Services.
- Good Experience in Tableau Analytics. Build Dashboards for clients using Tableau.
- Experience in end-to-end design and deploy rich Graphic visualizations with Drill Down and Drop-down menu option and Parameterized using Tableau.
- I am proficient in SQL databases **MSSQL Server, MySQL (RDBMS), Oracle DB, Postgres, and MongoDB**.

- Expert in various Azure services like Compute (Web Roles, Worker Roles), Caching, Azure SQL, NoSQL, Storage, and Network services, Azure Active Directory (AD), API Management, Scheduling, Azure Auto Scaling, and Azure shell, ARM, PowerShell Automation.

Technical Skills:

- **AWS Services:** S3, EC2, EMR, Redshift, RDS, Lambda, Kinesis, Glue, Athena, SageMaker, SNS, SQS, AMI, IAM, Cloud formation.
- **Databases:** Microsoft SQL Server, SQLite, MySQL, PostgreSQL, DB2, MongoDB, Cassandra, Redis.
- **Hadoop Components / Big Data:** HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, Yarn, Cloudera Manager, Kerberos, Pyspark Airflow, Kafka, Snowflake Spark Components.
- **Databases:** Oracle, Microsoft SQL Server, MySQL, DB2, Teradata, IBM DB2, MS SQL Server, Teradata, Snowflake.
- **Programming Languages:** Java, Scala, Impala, Python, PySQL.
- **Web Servers:** Apache Tomcat, WebLogic.
- **IDE:** Eclipse, Dreamweaver.
- **Methodologies:** Agile (Scrum), Waterfall, UML, Design Patterns, SDLC.
- **Currently Exploring:** Apache Flink, Drill, Tachyon.
- **Cloud Services:** AWS, Azure Data Factory / ETL / ELT / SSIS Azure Data Lake Storage Azure Data Bricks.
- **ETL Tools:** Talend Open Studio & Talend Enterprise Platform.
- **Reporting and ETL Tools:** Tableau, Power BI, AWS GLUE, SSIS, SSRS, Informatica, Data Stag.

Education:

Bachelor of Science in Mechanical engineering (2011-2015)

KKR&KSR INSTITUTE OF TECHNOLOGY AND SCIENCE

Professional Experience:

Verizon, NJ

Feb 2024 - Present

Lead Data Engineer

Responsibilities:

- Performed data analysis and developed analytic solutions. Data investigation to discover correlations / trends and the ability to explain them.
- Worked with Data Engineers, Data Architects, to define back-end requirements for data products (aggregations, materialized views, tables - visualization)
- Developed frameworks and processes to analyse unstructured information. Assisted in Azure Power BI architecture design.
- Designing and Developing **Oracle PL/SQL and Shell Scripts, Data Import/Export, Data Conversions and Data Cleansing.**
- Designed and optimized **PySQL** queries for data extraction and transformation in a distributed environment.
- Architect & implement medium to large scale BI solutions on Azure using Azure Data Platform services (Azure Data Lake, Data Factory, Data Lake Analytics, Stream Analytics)
- Performing data analysis, statistical analysis, generated reports, listings, and graphs using **SAS tools, SAS/Graph, SAS/SQL, SAS/Connect and SAS/Access.**
- Designed and developed Flink pipelines to consume streaming data from Kafka and applied business logic to massage and transform and serialize raw data.
- Leading the design and implementation of data pipelines using **PySpark** and AWS services such as **EMR, Glue, and Athena.**
- Authoring **Python (PySpark)** Scripts for custom UDF's for Row/ Column manipulations, merges, aggregations, stacking, data labelling and for all Cleaning and conforming tasks. Migrate data from on-premises to AWS storage buckets.
- Optimized transformation processes for Bloomberg data in **Snowflake** and **Redshift** using **SQL** and Python (**PySpark**).

- Developed **PySpark**-based ETL pipelines to process structured and unstructured data efficiently.
- Utilized **AWS SageMaker** for building and deploying machine learning models to enhance predictive analytics.
- Optimizing large-scale ETL workflows for performance and cost efficiency using **AWS Glue and PySQL**.
- Agile methodology includes test-driven and pair-programming concepts.
- Developed a Spark Streaming module for consumption of Avro messages from Kafka.
- Worked on Ingesting data by going through cleansing and transformations and leveraging AWS Lambda, AWS Glue and Step Functions.
- Created yaml files for each data source and including glue table stack creation. Worked on a python script to extract data from Netezza databases and transfer it to AWS S3
- Acquired and processed Bloomberg market data via Bloomberg Terminal, Bloomberg Data License (BDL), and Bloomberg APIs (B-PIPE) for real-time and historical data ingestion.
- Developed Lambda functions and assigned IAM roles to run python scripts along with various triggers (SQS, Event Bridge, SNS)
- Worked on Docker containers snapshots, attaching to a running container, removing images, managing Directory structures, and managing containers.
- Experienced in day-to-day DBA activities including schema management, user management (creating users, synonyms, privileges, roles, quotas, indexes, sequence), space management (table space, rollback segment), monitoring (alert log, memory, disk I/O, CPU, database connectivity), scheduling jobs, UNIX Shell Scripting.
- Developed complex Talend **ETL** jobs to migrate the data from flat files to database. Pulled files from main frame into Talend execution server using multiple ftp components.
- Developed ETL pipelines for transforming Bloomberg datasets, including financial statements, security prices, and economic indicators, into structured formats for analytics and reporting.
- Developed Talend **ESB** services and deployed them on **ESB** servers on different instances.
- Developed stored procedures/views in Snowflake and use in Talend for loading Dimensions and Facts.
- Developed merge scripts to UPSERT data into Snowflake from an **ETL** source.

Environment: Hadoop, Map Reduce, HDFS, Hive, Apache Flink, Spring Boot, Cassandra, Data Streaming, Data Lake, Sqoop, Oozie, SQL, PySQL, Kafka, Spark, Scala, GCP, GitHub, RedShift, Talend Big Data Integration, Solr, Impala

Citi Bank, Dallas, TX

Sep 2022 – Jan 2024

Sr Data Engineer

Responsibilities:

- Prepared the Estimations and Design specifications for the requirements by having initiative-taking discussions with the Business Analysts, Data Analysts and End users.
- Worked on Apache Hadoop ecosystem components like **MapReduce, HDFS, HBase, Hive, Sqoop, Pig, Oozie, Zookeeper, Flume, Spark, Python with CDH4&5 distributions and EC2 cloud computing with AWS**.
- Designed and maintained scalable ETL pipelines in **AWS Glue and PySpark** for processing high-volume transactional data.
- Analyzed source table and identify the best split key and optimum mapper counts for **sqoop**.
- Developed User Defined Function (UDF) to extend the functionality of Hive and Pig.
- Developed programs to convert embedded XML into JSON for Hive Table creation.
- Created External and Managed Hive tables and views from Json and Avro files.
- Design and Develop architecture for data curation.
- Implemented automated tools to do automated cluster to cluster copy.
- Identifying optimum partitioning and clustering to optimize data flow through cluster.
- Implemented Hive queries for data analysis to meet the requirements.
- Creation of Pig Latin Scripts to merge incremental data.
- Interact with business to gather requirements and propose effective and efficient design.

- Analyzed data which need to be loaded into Hadoop and contacted with respective source teams to get the table information and connection details.
- Used IBIS (in-house) Sqoop to import data from different RDBMS systems like Oracle, DB2 and Netezza and loaded into HDFS.
- Developed Oozie workflow schedulers to run multiple Hive and Pig jobs that run independently with time and data availability.
- Developed Data warehousing & Reporting solutions using **Informatica, Oracle Data Integrator (ODI), SSIS, Tableau, OBIEE, SSRS technologies.**
- Developed **PySQL** queries for complex data transformations and analytics.
- Performed administration activities like creating users, roles, responsibilities and backups for repository, topology. Purging the session logs and clean-up the stale sessions.
- Direct experience in visualizing the metrics data using Podium Data.
- Experience setting up and configuring Jenkins pipeline for Continuous testing and Continuous Deployments for all the non-prod and prod environments.
- Prepared detail migrations tasks for the code migration from Lower to Higher instances
- Troubleshooting the load performance, data issues and then providing detail root cause analysis
- Implemented Proof of Concept on the new solutions proposed for business requirements.

Environment: Cloudera Hadoop CDH5, AWS EC2&S3, Informatica, Oracle Data Integrator (ODI), Oracle BI, Podium Data, Looker, ERWIN, Python, Jenkins, Docker, Ansible, HDFS, Oracle, SQL Server, Flat Files, XMLs, SQL Server, Unix Scripting, Groovy, Python, Putty, SQL, PL/SQL, SAFe Agile Framework.

Johnson & Johnson, Bridgewater, NY

Nov 2021 – Aug 2022

AWS Data Engineer

Responsibilities:

- Developed strategy for cloud migration and implementation of best practices using AWS services like database migration service, AWS server migration service from On-Premises to cloud for large-scale commercial data workloads.
- Responsible for Setup and build AWS infrastructure using resources **VPC, EC2, S3, Dynamo DB, IAM, EBS, Route53, SNS, SES, SQS, Cloud Watch, Cloud Trail, Security Group, Auto scaling, and RDS using Cloud Formation templates.**
- Built reusable ETL pipelines using **AWS Glue** and **Lambda**, with orchestration through **AWS MWAA (Managed Workflows for Apache Airflow)** to process large volumes of commercial sales, marketing, and operations data.
- Developed ingestion workflows for **pharmaceutical product master data** and batch manufacturing records from **SAP** and **Oracle ERP** into S3-based data lake for downstream analytics.
- Led containerized microservices development and orchestration using **Docker, ECS, ECR, and Fargate** for scalable deployment of internal APIs and ETL workflows.
- Backed up AWS Post GREtoS3on daily job run on EMR using Data Frames and managed archiving in Glacier.
- Designed **ETL** processes to support **pharmaceutical regulatory reporting** such as **FDA** submissions and compliance checks with **21 CFR Part 11 standards.**
- Developed real-time analytics dashboards using **Athena and Quick Sight** to support executive-level commercial insights and campaign performance tracking.
- Created PySpark-based ETL workflows that supported transformation and normalization of raw operational data for BI reporting.
- Created Snowflake views to support analytics for **pharmacovigilance**, including adverse event detection and product safety monitoring dashboards.
- Worked on implementing data warehouse solutions in **AWS Redshift**, migrating datasets from various RDBMS systems and integrating them with **DynamoDB, RDS, ELB, and S3.**

- Partnered with data governance and compliance teams to ensure traceability and audit readiness of pipelines handling sensitive **healthcare and clinical data**.
- Automated metadata extraction and data lineage mapping using Python, reducing manual effort by over 70 hours monthly.
- Wrote scripts from scratch to build AWS infrastructure using Bash and Python, created Lambda functions for automated data ingestion and DynamoDB-S3 integration.
- Experience in **Infrastructure as Code (IaC)** using **Terraform** and **AWS CloudFormation**, including development of reusable modules for EC2, Redshift, and MWSA setup.
- Orchestrated and migrated CI/CD processes using Cloud Formation and Terraform, packer Templates and Containerized the infrastructure using Docker, which was setup in **OpenShift, AWS and VPCs**.
- Created Python scripts to handle PL/SQL constructs like cursors and loops unsupported by Snowflake, optimizing SQL logic for analytics use cases.
- Implemented cost dashboards on Snowflake resource usage and performance tuning in **QlikView** for internal stakeholders.
- Developed and deployed **ML** models in **Sage Maker** for **drug response prediction** based on **clinical attributes** and **lab data**.
- Utilized **EC2, CloudWatch**, and CloudTrail to monitor compute-intensive jobs, scale instances, and optimize cost performance tradeoffs across production and UAT environments.
- Worked on **Apache Spark** to enhance the performance and optimization of existing algorithms, leveraging Spark SQL and YARN for data transformations and joins across large HDFS datasets.

Environment: Spark, Hive, Pig, Spark SQL, Spark Streaming, HBase, Sqoop, Kafka, AWS EC2, S3, Glue, Lambda, Redshift, DynamoDB, MWSA, Cloudera, Python, Snowflake, QlikView, JSON, OpenShift, AWS Glacier, Terraform, Docker, EMR, Bash, CloudFormation, ECS, ECR, Fargate, CloudTrail, SageMaker.

Niche Soft INC, India

Apr 2015 – Sep 2021

Data Engineer

Responsibilities:

- Designed and built **Spark/PySpark** based ETL pipelines for migration of credit card transactions, accounts, and customer data into enterprise Hadoop Data Lake. Developed strategies in handling large datasets using partitions, Spark SQL, broadcast joins and performance tuning.
- Built and implemented performant data pipelines using **Apache Spark** on **AWS EMR**. Performed maintenance of data integration programs into Hadoop and RDBMS environments from both structured and semi- structured data source systems.
- Developed scalable ETL pipelines using **Spark, PySpark, and AWS Glue** to process structured and unstructured data.
- Automated data ingestion from multiple sources into a centralized data lake using **Athena and Glue**.
- Developed performance tuning on existing Hive queries and UDF's to analyze the data. Used Pig to analyze datasets and perform transformation according to requirements.
- Supervised data profiling and data validation to ensure the accuracy of the data between the source and the target systems. Performed job scheduling and monitoring using Auto sys and quality testing using ALM tool.
- Worked on building Tableau desktop reports and dashboards to report customer data.
- Built and published customized interactive Tableau reports and dashboards along with data refresh scheduling using Tableau Desktop.
- Snowflake - data warehouse to consume the data from C3 Platform.
- Involved in S3 event notifications, an SNS topic, an SQS queue, and a Lambda function sending a message to the Slack channel.
- Transformed **Teradata** scripts and stored procedures to SQL and Python running on Snowflake's cloud platform.

- Deploy and monitor scalable infrastructure on Amazon web services (AWS) and configuration management instances and Managed servers on the Amazon Web Services (AWS) platform using Ansible configuration management tools and Created instances in AWS as well as migrated data to AWS from data Centre.
- Automated tasks of extracting metadata and lineage from tools using Python scripts and saved 70+ hours' manual efforts.
- Analyzed the system requirement specifications and in client interaction during requirements specifications.
- Providing daily reports to the Development Manager and participating in both the design phase and the development phase. Utilized Agile Methodology and SCRUM Process

Environment: AWS, Hadoop, Python, Pyspark, SQL, Snowflake, Data bricks/Delta Lake, AWS S3, AWS Athena and AWS EMR.